

Forecasting 311 Rat Sightings in New York City

Github: <https://github.com/ErDOS-Projects/spring-2026-rat-activity-nyc>

Datasets:

<https://open-meteo.com/en/docs>

https://data.cityofnewyork.us/Social-Services/Rat-Sightings/3q43-55fe/about_data

Background and Project Overview

This project analyzes New York City (NYC) 311 rat sighting data to forecast daily rat sightings with 14 day forecast horizons at both the citywide and borough levels. The team's primary goal was to determine if one can predict the number of daily 311 rat sightings 14 days horizons into the future, and whether predictions improve when weather information¹ and engineered features are added. Our analysis showed clear yearly seasonal behavior, with sightings increasing in summer months and decreasing in winter months. We also observed² trend changes over time, including growth in sightings through 2020 to 2022 followed by decline beginning in 2023. Since 311 rat sighting calls are made by people³, we expected human behavior to play a role and observed that holidays and weekends had a significant impact. For example, we saw a drop in rat sightings reported in the days leading up to a federal holiday and an increase in the following days. Overall, we saw that reporting is shaped by day of week, time of year, weather, and human events, making short-term forecasting practical.

Stakeholders

Given the nature of the project, our primary stakeholders are the NYC Department of Health and Mental Hygiene and residents of NYC.

Data Obtainment, Cleaning, and Feature Engineering

We grouped data based on the "Created Date" of each report and the spatial level we were interested in. When grouping, we were careful to insert missing dates and assign "0" for the number of rat sightings reported if no reports were found. We focused on data from 2020-01-01 to 2026-02-28 inclusive. We used [open-meteo](https://open-meteo.com/en/docs) for weather data. We engineered features that felt relevant based on domain knowledge and experimentation. For example, we added the columns listed below.

- Columns corresponding to day of the year, day of the week, month, and week of the year
- Columns with max/min/mean apparent temperature and measurements of precipitation and snowfall
- Columns containing lags of the previous features (we are careful to use lags of at least 14 days to avoid temporal leakage) and lags of the target (which is the daily number of sightings)

Modeling Approach

¹ <https://www.science.org/doi/10.1126/sciadv.ads6782>

² Also verifiable from an STL decomposition.

³ <https://portal.311.nyc.gov/about-nyc-311/>

We held out 2025-03-01 to 2026-02-28 as a final evaluation set. We had a list of models to consider and depending on the best performer, we considered derivative models or ensembles of models. Our evaluation process was to take 2020-01-01 to 2025-02-28 and use a walk-forward cross validation scheme with 26 folds and size 14 test sets. We took our main KPI to be the mean RMSE over the 26 folds. We chose mean RMSE since we wanted overall performance to be good, and to have sensitivity to outliers since the data showed occasional outliers in rat sightings. For a secondary metric, we chose mean MAPE. The models we considered were the seasonal average, Holt-Winters, SARIMA, XGBoost, and Prophet models. Models built on top of these such as NeuralProphet and Prophet + XGBoost on residuals (referred to as the “hybrid model” here on) were also considered based on performance of the first five.

Results

For individual boroughs as well as citywide, the best-performing models were either Prophet, NeuralProphet, or hybrid models incorporating Prophet. At the citywide level, for instance, the seasonal average model had a mean RMSE of 17.57 while Prophet had a mean RMSE of 13.08. The hybrid model where we used XGBoost to predict the residuals reduced the mean RMSE to 12.14. On the holdout set and using the best hyperparameters and features identified through Optuna by that point, the hybrid model had a mean RMSE of 10.98.

Feature importance analysis from XGBoost showed that weather-related features were among the most important predictors. At the borough level, Prophet and Prophet-based models consistently outperformed alternatives such as SARIMA and Holt-Winters. This aligns with EDA where we observed that the data had strong yearly and weekly seasonality, holiday effects, and trend changes, and these are patterns Prophet is designed to capture.

Limitations and Future Work

In the future, we would like to incorporate additional features and use the other datasets we had included to improve performance. We could also consider the multivariate time series forecasting with daily rat sightings for each borough as target variables and use [tools](#) designed for it. For citywide forecasting, we might also choose to forecast by borough and add up the predictions. For city administrators, it could be helpful to predict weekly average per day sightings. Weekly averaging would smooth away the influence of outliers in the data. We had considered this and would like to continue in this direction. Other additions to our work are to change from point forecasts to giving prediction intervals and provide more interpretability for our models. Another idea would be to change the granularity of the problem and consider modeling at the ZIP code level. We observed that the volume of 311 calls were concentrated in a handful of ZIP codes.⁴ It would be useful for stakeholders to forecast at this spatial level. One idea might be to try using [Graph Neural Networks](#), and this is left for future work.

⁴ Between 2020 and the end of 2025, zip code 11697 had zero rat sightings reported while zip code 11385 had nearly three thousand.